

Infosys Springboard Virtual Internship 6.0

Completion Report

Team Details

Batch Number: Batch13

Start date: 25/02/2026

Names: Srihitha Mandaji

Internship Duration: 8 Weeks

1. Project Title

FarmChainX – AI Driven Agricultural Traceability Network

2. Project Objective

The objective of the FarmChainX project is to develop a transparent and user-friendly platform that enables end-to-end traceability of agricultural products from farmers to consumers. The system aims to ensure product authenticity and build trust by allowing users to track the complete journey of each product using QR code technology. It also focuses on enabling farmers to manage their crops efficiently, helping distributors handle logistics operations smoothly, and allowing consumers to verify and purchase products with confidence. Additionally, the platform improves communication and visibility across all stakeholders, ultimately enhancing efficiency, accountability, and overall quality in the agricultural supply chain.

3. Project description in detail

The proposed model introduces FarmChainX as an AI-driven agricultural traceability system designed to address the limitations of traditional supply chain methods. It provides farmers, distributors, and consumers with a structured and user-friendly platform to manage and track agricultural products efficiently. Each product is assigned a unique QR code and stored in a centralized system, ensuring that every stage of the product's journey is recorded and easily accessible. This organized approach improves transparency, reduces chances of data loss or manipulation, and builds trust among all stakeholders.

Traceability is an essential component of any modern supply chain system as it enables clear visibility and accountability at every stage. In this system, farmers can register and manage crop details, distributors can handle transportation, storage, and delivery operations, and consumers can verify product authenticity by scanning QR codes. This ensures effective communication and coordination between all participants in the supply chain. An efficient traceability system also helps in early identification of issues such as delays, spoilage, or inconsistencies, allowing timely action to be taken. By providing complete product history and real-time insights, the system enhances reliability, improves decision-making, and ensures better quality assurance in the agricultural ecosystem.

4. Timeline Overview

Week	Activities Planned	Activities Completed
Week 1	Use case diagram, Workflow diagram, ER diagram, Feature diagram, Table Structure	Use case diagram, Workflow diagram, ER diagram, Feature diagram, Table Structure
Week 2	UI/UX Design of User Login, Admin Login, Staff Login Page, User Dashboard	UI/UX Design of User Login, Admin Login, Staff Login Page, User Dashboard
Week 3	Frontend development of User Login Page, Admin Login Page, Staff Login Pages and User Dashboard Pages	Frontend development of User Login Page, Admin Login Page, Staff Login Pages and User Dashboard Pages
Week 4	Working on Database of the Project, creating tables, Working on Database tables using MySQL	Working on Database of the Project, created tables, Working on Database tables using MySQL
Week 5	Working on the Backend Codes, Structures of Files in Backend and Frontend, Designing navigation and Flow of the Website	Worked on the Backend Codes, Structures of Files in Backend and Frontend, Designed the navigation and Flow of the Website
Week 6	Testing the working of the codes with the help of Postman, working on codes, Improvisation of code quality, Integration of Login Pages	Tested the working of the codes with the help of Postman, worked on codes, improvised code quality, Integrated the Login Pages
Week 7	Working on Frontend pages such as profile page and logout page making them improved, working on navigation through dashboards and complete functionality	Worked on Frontend pages such as profile page and logout page making them improved, worked on navigation through dashboards and completed functionality of the dashboards to work well
Week 8	Completion of Project and Presentation of The Project	Completed the Project, Presented the Project

5a. Key Milestones

Milestone	Description	Date Achieved
Project Kickoff	Use case diagram, Workflow diagram, ER diagram, Feature diagram, Table Structure, UI/UX Design of User Login, Admin Login, Staff Login Page, User Dashboard	25 February 2026
Prototype/First Draft	Frontend development of User Login Page, Admin Login Page, Staff Login Pages and User Dashboard Pages	16 March 2026
Mid-Term Review	Worked on Frontend pages such as profile page and logout page making them improved, worked on navigation through dashboards and completed functionality of the dashboards to work well	25 March 2026
Final Submission	Completed the Project	02 April 2026
Presentation	Presented the Project to the Panel members and batch mates	

5b. Project execution details**Step 1: User Registration**

The process begins when a new user opens the registration page.

The user selects their role as Farmer, Consumer, or Distributor and enters basic details such as name, email, and password.

The system validates whether all required fields are filled correctly.

If the details are valid, the information is stored in the system, and the user is successfully registered.

Step 2: User Login

After registration, the user proceeds to the login page.

The user enters their credentials such as email and password.

The system verifies these details with the stored data.

If the credentials are correct, the user is logged in successfully and granted access to the system.

Step 3: Role – Based Access

Once logged in, the system identifies the user's role.

Based on the role (Farmer, Consumer, or Distributor), the system provides access to specific features.

Each user sees a customized dashboard according to their responsibilities.

Step 4: Farmer – Product Entry

The farmer accesses the dashboard and adds crop or product details. Information such as crop name, quantity, and other relevant details are entered. The system validates the data and stores it securely. Each product is recorded as a unique entry in the system.

Step 5: QR Code Generation

After adding product details, the system generates a unique QR code for each product. This QR code acts as a digital identity and is linked to the product's data. It is used throughout the supply chain for tracking and verification.

Step 6: Distributor – Logistics Handling

The distributor logs into the system and views assigned products. They manage operations such as transportation, storage, and delivery. The system updates product status at each stage of movement. All activities are recorded to maintain transparency.

Step 7: Tracking and Traceability

At every stage, the product journey is tracked in the system. Details such as location, status, and handling information are updated. This ensures complete traceability from farm to consumer.

Step 8: Consumer Interaction

The consumer logs into the system and explores available products. They can scan the QR code of a product to verify its authenticity. The system displays the complete product history, including origin and supply chain details. Consumers can also add products to cart and place orders.

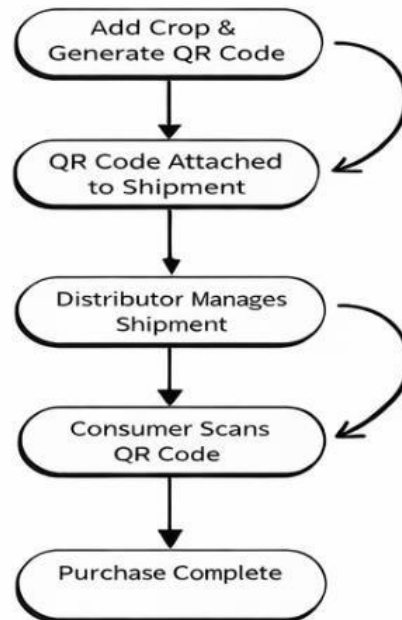
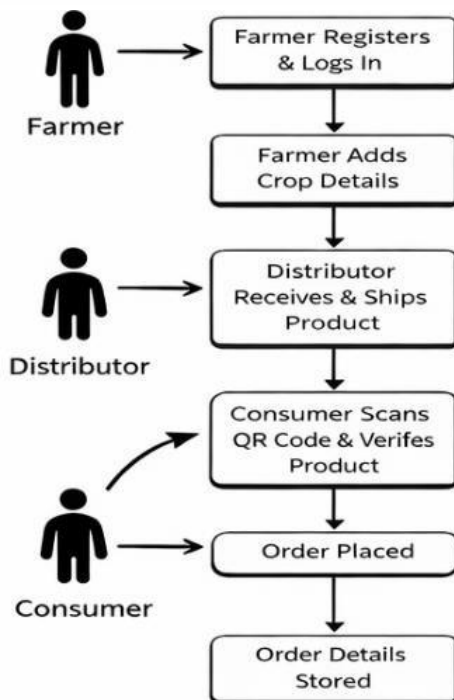
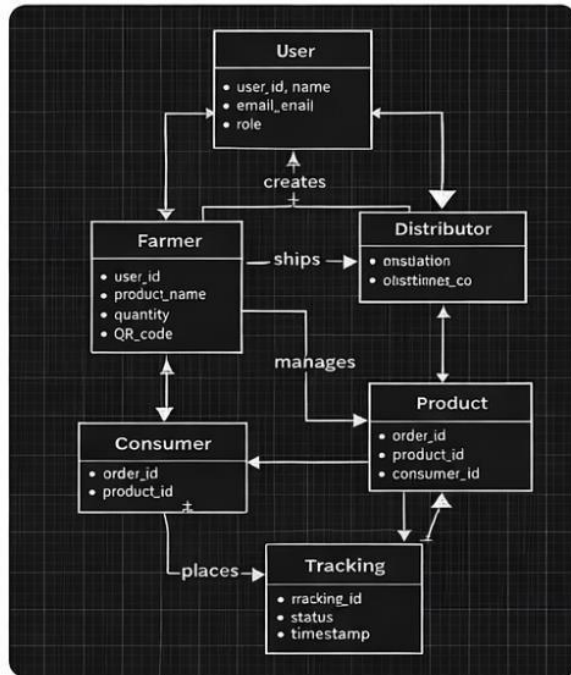
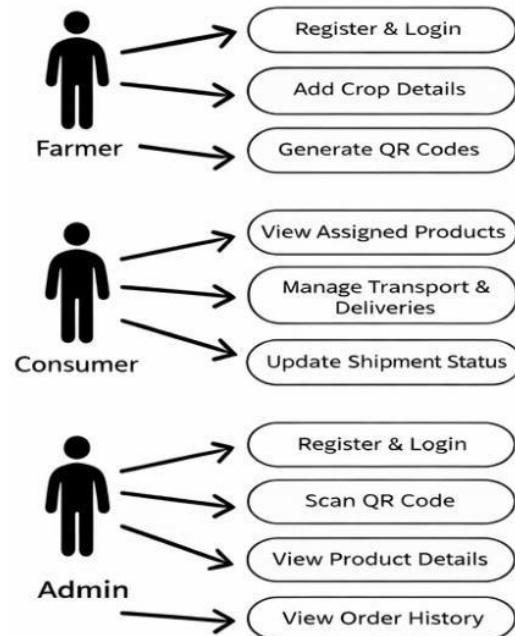
Step 9: Order Management

Once a product is purchased, the order details are stored in the system. Consumers can view their order history and track purchased products. This ensures proper record management and easy access to previous transactions.

Step 10: Monitoring and Validation

The system ensures that all actions are validated at each step. Only authorized users can access specific functionalities. All data is securely stored and updated in real time. This ensures reliability, accuracy, and smooth system performance.

6. Snapshots / Screenshots



UI/UX DESIGNS

FARMTRACE CONNECT

TRACE. TRUST. TRANSFORM.

AI-powered agricultural traceability for every farm, distributor, and consumer. See every harvest, shipment, and quality checkpoint in one polished experience.

Farmer

Manage crops, record harvests, track production.

PRODUCER

Consumer

Trace origins, verify quality, make informed choices.

END USER

Get Started

Already have an account? [Sign in](#)

REGISTER

Farmer onboarding

Create an account for FarmChainX and start managing your supply chain with rich insights, traceability, and quality monitoring.

FULL NAME

Your name

E MAIL

you@farm.io

PASSWORD

Create a password

Create Farmer account

Already have an account? [Log in](#)

FARMCHAINX

Sign in to your portal

Access your role-specific command center for farmers, consumers, and administrators.

ROLE

Farmer

Login details

Enter your workspace credentials to continue.

N A M E

Your name

N A M E

Your name

E M A I L

you@farm.io

P A S S W O R D

Continue

Not registered? [Create an account](#)

FARMCHAINX
PRODUCER

Dashboard

Add Crop

My Crops

Payments

SIGNED IN AS
ramya

Sign out

SYSTEM ONLINE

Farmer Dashboard

Overview of your farm operations and supply chain

ACTIVE CROPS

5

+1 this month

SHIPMENTS

5

5 in transit

REVENUE

₹14,600

+18%

QUALITY

99%

+2 pts

RECENT CROPS

rice

FCX-4053 - ...

Growing

2026-04-15

wheat

FCX-5354 - ...

Growing

2026-04-13

wheat

FCX-8113 - ...

Harvested

2026-04-12

CROP DETAILS

Add Crop

Create a new crop record and store harvest inventory details.

Crop Name *

e.g. Organic Wheat

Quantity *

e.g. 500 kg

Price per Unit (Optional)

e.g. 50

Harvest Date *

dd-mm-yyyy

Location *

Farm location

Quality (%)

e.g. 85

Enter a numerical quality score (0-100).

SYSTEM ONLINE

Farmer Dashboard

Overview of your farm operations and supply chain

My Crops

ID	NAME	TYPE	QTY	DATE	LOCATION	QUALITY	STATUS
FCX-4053	rice		200	2026-04-15	hyd	69%	Growing
FCX-5354	wheat		250	2026-04-13	hyd	79%	Growing
FCX-8113	wheat		259	2026-04-12	hyd	68%	Harvested
FCX-1203	Wheat	Grain	1,800 kg	2026-03-15	Madhya Pradesh	--%	Growing
FCX-0847	Organic Rice	Grain	2,400 kg	2026-02-28	Punjab Farm	--%	Harvested

PAYMENTS

Recent payments

Total revenue: ₹14,600

COMPLETED

Seed purchase for winter season

₹7,800

2026-02-10

Transport advance for shipment

₹4,200

2026-02-22

PENDING

Packaging material payment

₹2,600

2026-03-02

FARMCHAINX
END USER

Dashboard

Marketplace

Trace Product

My Purchases

SIGNED IN AS
ramya

Sign out

SYSTEM ONLINE

Consumer Dashboard

Discover verified products and trace their journey

TRACED

1

+this week

VERIFIED

22

91.6% rate

AVG SCORE

93

+2 pts

PURCHASES

6

This month

Marketplace

Search for products or farmers...

Fresh Tomatoes

Farmer: Ram Kumar

₹45/kg

Review: 4.5 / 5

Quality: 92%

Organic

Organic Carrots

Farmer: Vikram Singh

₹50/kg

Review: 4.7 / 5

Quality: 95%

Organic

Green Peppers

Farmer: Aarti Patel

₹60/kg

Review: 4.2 / 5

Quality: 89%

Organic

Basmati Rice

Farmer: Suresh Rao

₹90/kg

Review: 4.8 / 5

Quality: 97%

Fresh Spinach

Farmer: Meena Kumari

₹30/bunch

Review: 4.1 / 5

Quality: 85%

Organic

Red Onions

Farmer: Ajay Sharma

₹35/kg

Review: 4 / 5

Quality: 80%

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₹30/bunch

Review: 4.1 / 5

Quality: 85%

Organic

Sweet Corn

Farmer: Sunita Devi

Potatoes

Farmer: Ravi Kumar

Your Cart

×

Fresh Spinach

Meena Kumari

₹30 x 1

Remove

Organic Carrots

Vikram Singh

₹50 x 1

Remove

Green Peppers

Aarti Patel

₹60 x 1

Remove

Proceed to Pay

Order History (9)

ID	Product	Qty	Total	Date	Status	Rating
ORD-001	Fresh Tomatoes	1 kg	₹45	2026-04-13	Delivered	★★★★★
ORD-002	Green Peppers	1 kg	₹60	2026-04-13	Delivered	★★★★★
ORD-003	Green Peppers	1 kg	₹60	2026-04-13	Delivered	★★★★★
ORD-004	Organic Carrots	1 kg	₹50	2026-04-13	Delivered	★★★★★
ORD-005	Fresh Tomatoes	1 kg	₹45	2026-04-13	Delivered	★★★★★
ORD-006	Organic Carrots	1 kg	₹50	2026-04-13	Delivered	★★★★★
ORD-007	Fresh Spinach	1 bunch	₹30	2026-04-20	Delivered	★★★★★
ORD-008	Organic Carrots	1 kg	₹50	2026-04-20	Delivered	★★★★★
ORD-009	Green Peppers	1 kg	₹60	2026-04-20	Delivered	★★★★★

Camera Scanner

Tap to scan a QR code with your camera.

fcx-a1b2

Scan

Organic Wheat

Scanned

Code: FCX-A1B2 Origin: Punjab Farm

Farmer: John Farmer Quantity: 500 kg

Harvest Date: Feb 15, 2026 Scanned At: 6:35:07 pm

AI Quality Analysis

92%

Quality Score

Risk Assessment

✓ Contamination Risk

No contaminants detected

Product scanned!
Organic Wheat added to scanned list

Screenshots of project Output: Login Page

REGISTER

Farmer onboarding

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FULL NAME

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PASSWORD

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REGISTER

Consumer onboarding

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FULL NAME

E MAIL

PASSWORD

Create Consumer account

Already have an account? [Log in](#)

FARMCHAINX

Sign in to your portal

Access your role-specific command center for farmers, consumers, and administrators.

ROLE

Farmer

Login details

Enter your workspace credentials to continue.

NAME

NAME

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PASSWORD

Continue

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User Dashboard

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PRODUCER

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Add Crop

My Crops

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SIGNED IN AS
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ACTIVE CROPS
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+1 this month

SHIPMENTS
5
5 in transit

REVENUE
₹14,600
+18%

QUALITY
99%
+2 pts

RECENT CROPS

rice
FCX-4053 - ...
Growing
2026-04-15

wheat
FCX-3354 - ...
Growing
2026-04-13

wheat
FCX-8513 - ...
Harvested
2026-04-12

FARMCHAINX
END USER

Dashboard

Marketplace

Trace Product

My Purchases

SIGNED IN AS
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SYSTEM ONLINE

Consumer Dashboard

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CROP DETAILS

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e.g. 50

Harvest Date *
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Location *
Farm location

Quality (%)
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Recent Payments Page

PAYMENTS

Recent payments

Total revenue: ₹14,600

COMPLETED

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Market Place Page

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Basmati Rice

Farmer: Suresh Rao

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Review: 4.8 / 5

Quality: 97%

Fresh Spinach

Farmer: Meena Kumari

₹30/bunch

Review: 4.1 / 5

Quality: 85%

Organic

Red Onions

Farmer: Ajay Sharma

₹35/kg

Review: 4 / 5

Quality: 80%

pg. 12

Cart Items Page

Marketplace

Search for products or farmers...

Fresh Tomatoes
Farmer: Ram Kumar
₹45/kg
Review: 4.5 / 5
Quality: 92%
Organic

Organic Carrots
Farmer: Vikram Singh
₹50/kg
Review: 4.7 / 5
Quality: 95%
Organic

Basmati Rice
Farmer: Suresh Rao
₹90/kg
Review: 4.8 / 5
Quality: 97%

Fresh Spinach
Farmer: Meena Kumari
₹30/bunch
Review: 4.1 / 5
Quality: 85%
Organic

Sweet Corn
Farmer: Sunita Devi

Potatoes
Farmer: Ravi Kumar

Your Cart

Fresh Spinach
Meena Kumari
₹30 x 1 Remove

Organic Carrots
Vikram Singh
₹50 x 1 Remove

Green Peppers
Aarti Patel
₹60 x 1 Remove

Proceed to Pay

Order History Page

Order History (9)						
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ORD-004	Organic Carrots	1 kg	₹50	2026-04-13	Delivered	★★★★★
ORD-005	Fresh Tomatoes	1 kg	₹45	2026-04-13	Delivered	★★★★★
ORD-006	Organic Carrots	1 kg	₹50	2026-04-13	Delivered	★★★★★
ORD-007	Fresh Spinach	1 bunch	₹30	2026-04-20	Delivered	★★★★★
ORD-008	Organic Carrots	1 kg	₹50	2026-04-20	Delivered	★★★★★
ORD-009	Green Peppers	1 kg	₹60	2026-04-20	Delivered	★★★★★

Traceability Page

Camera Scanner

Tap to scan a QR code with your camera.

fcx-a1b2

Scan

Organic Wheat

Scanned

Code: **FCX-A1B2** Origin: **Punjab Farm**
Farmer: **John Farmer** Quantity: **500 kg**
Harvest Date: **Feb 15, 2026** Scanned At: **6:35:07 pm**

AI Quality Analysis

92%
Quality Score

Risk Assessment

✓ Contamination Risk
No contaminants detected

Product scanned!
Organic Wheat added to scanned list

7. Challenges Faced

I faced challenges while implementing role-based access and ensuring proper flow between different modules like farmer, consumer, and distributor. Maintaining consistent data handling across these modules and debugging logical errors was a bit difficult. We also faced issues in aligning different components and ensuring smooth navigation between features. However, resolving these challenges helped us improve our understanding of system design and integration.

8. Learnings & Skills Acquired

Languages In Frontend: HTML, CSS, JavaScript

Languages In Backend: Java Spring Boot

Database: MySQL

For Designing: Adobe Express

For Images: Draw.io, Adobe Express, PowerPoint

For UI/UX Designing: Canva, Figma, Adobe Express

Repository: Git, Git Hub

Presentation: PowerPoint Presentation

Testing: Manual/ System testing

9. Testimonials from team

I gained a lot of hands on experience on how to design the way we work on a project and the importance of the diagrams that I had earlier learnt in Software engineering the use case diagram, the workflow diagram, the sequence diagram, and the ER diagram these diagrams helped me get a perspective on what I was building and how the workflow is to be designed from that.

10. Conclusion

The internship was a beginner friendly atmosphere for me to learn, understand things, become consistent, be active and present my work with all the confidence that I have gained throughout my journey and get to know the importance of how a structured and well planned work distribution can help us deeply in understanding a lot of different factors that would support us in the successful creation of a complete project that could be useful to others. Also encourage us to build more of what we face daily and turn a problem into a project that helps everyone that faces the problem.

11. Acknowledgements

I would like to extend my gratitude and thank Sandeep sir for his constant support and motivation, his suggestions and feedback for every presentation we do and solving all the questions we had in our mind throughout these 8 weeks of training. I would like to sincerely thank him for his consistent guidance and mentorship towards each one of us. I would like to thank Infosys Springboard for providing this amazing opportunity to students across India empowering everyone across the country.